

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

temp.py X

spyder01.py X

spyder02.py X

spider03.py X

mlr_sklearn_statsmodel.py X

Poly_regression.py X

1

#Linear regression

2

import numpy as np

3

import pandas as pd

4

import matplotlib.pyplot as plt

5

6

dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\emp_sal.csv")

7

8

x = dataset.iloc[:,1:2].values

9

y = dataset.iloc[:,2].values

10

11

from sklearn.linear_model import LinearRegression

12

lin_reg = LinearRegression()

13

lin_reg.fit(x,y)

14

15

plt.scatter(x, y, color='red')

16

plt.plot(x, lin_reg.predict(x), color = 'blue')

17

plt.title('Linear Regression Graph')

18

plt.xlabel('Position Level')

19

plt.ylabel('Salary')

20

plt.show()

21

22

#Polynomial regression

23

lin_model_pred = lin_reg.predict([[6.5]])

24

print(lin_model_pred)

25

26

from sklearn.preprocessing import PolynomialFeatures

27

poly_reg = PolynomialFeatures(degree=5)

28

x_poly = poly_reg.fit_transform(x)

29

30

poly_reg.fit(x_poly,y)

31

32

lin_reg_2 = LinearRegression()

33

lin_reg_2.fit(x_poly,y)

34

35

plt.scatter(x, y, color = 'red')

36

plt.plot(x, lin_reg_2.predict(poly_reg.fit_transform(x)), color = 'blue')

37

plt.title('Truth or bluff (polynomial Regression)')

38

plt.xlabel('Positional level')

39

plt.ylabel('Salary')

40

plt.show()

C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\Poly_regression.py

Name

Type

dataset

DataFrame

lin_model_pred

Array of float64

lin_reg

linear_model_base.LinearRegression

lin_reg_2

linear_model_base.LinearRegression

poly_model_pred

Array of float64

poly_reg

preprocessing_polynomial.PolynomialFeatures

x

Array of int64

x_poly

Array of float64

y

Arrav of int64

Help

Variable Explorer

Plots

Files

Console 1/A X

Python 3.12.7 | packaged by Anaconda, Inc. | (main, Oct 4 2024, 13:17:27) [MSC v.1929 64 bit (AMD64)]

Type "copyright", "credits" or "license" for more information.

IPython 8.27.0 -- An enhanced Interactive Python.

Restarting kernel...

In [1]:

conda (Python 3.12.7)

Completions: conda

LSP: Python

Line 26, Col 43

ASCII

CRLF

RW

Mem 85%

Spyder (Python 3.12)

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C:\Users\Lenovo\OneDrive\Desktop\Python\spyder

C:\Users\Lenovo\OneDrive\Desktop\Python\spyder\Poly_regression.py

```
1 #Linear regression
2 import numpy as np
3 import pandas as pd
4 import matplotlib.pyplot as plt
5
6 dataset = pd.read_csv(r"C:\Users\Lenovo\Downloads\
7
8 x = dataset.iloc[:,1:2].values
9 y = dataset.iloc[:,2].values
10
11 from sklearn.linear_model import LinearRegression
12 lin_reg = LinearRegression()
13 lin_reg.fit(x,y)
14
15 plt.scatter(x, y, color='red')
16 plt.plot(x, lin_reg.predict(x), color = 'blue')
17 plt.title('Linear Regression Graph')
18 plt.xlabel('Position Level')
19 plt.ylabel('Salary')
20 plt.show()
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22 #Polynomial regression
23 lin_model_pred = lin_reg.predict([[6.5]])
24 print(lin_model_pred)
25
26 from sklearn.preprocessing import PolynomialFeatures
27 poly_reg = PolynomialFeatures(degree=6)
28 x_poly = poly_reg.fit_transform(x)
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39 plt.ylabel('Salary')
```

Linear Regression Graph

Position Level	Salary (1e6)
1	0.05
2	0.05
3	0.06
4	0.08
5	0.11
6	0.15
7	0.20
8	0.30
9	0.50
10	1.00

Help Variable Explorer Plots Files

Console 1/A X

```
575694.63869464 950461.53846153]
In [4]: runfile('C:/Users/Lenovo/OneDrive/Desktop/Python/spyder/Poly_regression.py', wdir='C:/Users/Lenovo/OneDrive/Desktop/Python/spyder')
```

IPython Console History

conda (Python 3.12.7) Completions: conda LSP: Python Line 27, Col 39 ASCII CRLF RW Mem 86%

Spyder (Python 3.12)

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133 %

Salary

1e6

Truth or bluff (polynomial Regression)

Positional level

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Spyder (Python 3.12)

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Truth or bluff (polynomial Regression)

Positional level

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133 %

Salary

1e6

Truth or bluff (polynomial Regression)

Positional level

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Console 1/A X

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